

Drone Payload Structure Design and Analysis

In this project, you are optimizing drone arm design, including geometry and material selection, to maximize payload capacity while ensuring structural integrity under motor-induced loads. You will use MATLAB to perform force balance analysis and finite element modeling to quantitatively support your design solution.

Use this Live Script as a template to get started on your project solution. You may include helper functions at the bottom of this document or as separate files. For a tutorial on how to use Live Scripts, check out this [video](#).

Suggested Tasks

1. Gather your starter assumptions and load the material properties into MATLAB.
2. Propose at least two drone arm designs
3. Perform a thrust-to-weight analysis to evaluate each of your designs across all material options. Your design must meet a minimum payload requirement and safety standards.
4. Perform finite element analysis across all design and material options.
5. Recommend a final design solution.

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Break Down the Problem

In the space below, define the problem statement:

The objective is to select a drone arm geometry and material combination that maximizes payload capacity while satisfying the minimum thrust-to-weight ratio and maintaining an acceptable structural safety factor under motor loading.

List the requirements, constraints, and success criteria for the project:

- The drone must support a minimum payload of 0.5 kg.
- The drone must meet a thrust-to-weight ratio of at least 2:1.
- The quadcopter configuration uses 4 motors and 4 propellers.
- Each motor can provide up to 1 kg of thrust.
- The total mass of the drone without arms or payload is 1 kg.
- The design must use at least two distinct arm geometries.
- The arm design must be evaluated across the available material options.
- The final selected design must satisfy the required payload and flight safety constraints.

- The structural design must maintain an acceptable factor of safety, with the target engineering range taken as approximately $\sim 1.5\text{-}2+$.
- Material cost should be considered as an optional optimization criterion in addition to the required performance and safety checks.

What is your proposed solution?

- The proposed solution is to compare two distinct drone arm geometries and evaluate them across the available material options using thrust-to-weight analysis and a single-arm finite element model to identify the combination that maximizes payload capacity while satisfying the required flight and structural safety constraints.

List the steps you will need to take to achieve your proposed solution:

1. Gather Starter Assumptions and Load Properties into MATLAB
2. Develop at Least Two Distinct Drone Arm Designs in CAD or by Hand
3. Write a MATLAB Function to Perform a Thrust-to-Weight Analysis Across All Design and Material Options
4. Create a 3D Model of One Drone Arm Using CAD Software of Your Choice
5. Import Your Model into MATLAB and Perform FEA
6. Report the Overall Maximum Displacement, Maximum Stress, and Factor of Safety for Each Design and Material Combination, Including Visualizations of the X-, Y-, and Z-Displacement and von Mises Stress
7. Recommend a Final Drone Arm Design That Meets All Required Constraints

What challenges do you face in the process of achieving the solution?

- The analysis is simplified to a single-arm structural model rather than the full four-arm drone assembly.
- Material properties are treated as homogeneous and isotropic, which is an approximation for composite and wood materials.
- The arm geometry and loading conditions are idealized rather than fully representing the complete flight environment.
- The FEA model captures only the dominant motor thrust and motor weight loading, while other real-world effects such as vibration, landing loads, and aerodynamic forces are not included.
- The design must balance payload capacity, structural stiffness, and material cost under the stated assumptions.

Task 1: Gather your starter assumptions and load material properties

Load the provided `droneArmMaterials.mat` file into MATLAB. This contains a data struct with the material properties for six different materials commonly used for drones (carbon fiber composite, aluminum alloy, fiberglass composite, PLA plastic, ABS plastic, and wood). For each material, you have been provided the following information:

- Density (ρ), in $\frac{\text{kg}}{\text{m}^3}$

- Young's Modulus (E) in Pa
- Poisson's ratio (ν), unitless
- Yield strength, in Pa
- Average cost per meter in US dollars

For this project, you can assume the following design parameters:

- Quadcopter drone with 4 motors and 4 propellers
- Each motor can provide up to 1kg thrust
- Acceleration due to gravity: $g = 9.81 \frac{m}{s^2}$
- Total weight of the drone components without the arms or the payload: 1kg

The breakdown of the weight of each drone component is as follow:

- Battery = 0.3kg
- Each motor = 65g, 4 motors = 0.26kg
- Each propeller = 10g, 4 propellers = 0.04 kg
- Each electronic speed controller = 30g, 4 electronic speed controllers = 0.12 kg
- Flight controller and electronics = 0.05kg
- Receiver/GPS = 0.03kg
- Drone frame (center body) excluding arms = 0.2kg

```
clc; clear; close all;

% Load material properties for the six drone arm materials
load droneArmMaterials.mat;

% Define drone component masses (kg)
batteryMass = 0.30; % Battery
motorMass = 65 / 1000; % Each motor
numberOfMotors = 4; % Number of motors
totalMotorMass = motorMass * numberOfMotors;

propellerMass = 10 / 1000; % Each propeller
numberOfPropellers = 4; % Number of propellers
totalPropellerMass = propellerMass * numberOfPropellers;

speedControllerMass = 30 / 1000; % Each electronic speed controller
numberOfSpeedControllers = 4; % Number of electronic speed controllers
totalSpeedControllerMass = ...
    speedControllerMass * numberOfSpeedControllers;

flightControllerMass = 0.05; % Flight controller and electronics
receiverGPSSMass = 0.03; % Receiver/GPS
droneFrameMass = 0.20; % Drone frame excluding arms
```

```

% Calculate total mass of the drone without arms or payload
totalDroneMassWithoutArmsOrPayload = ...
    batteryMass + ...
    totalPropellerMass + ...
    totalMotorMass + ...
    totalSpeedControllerMass + ...
    flightControllerMass + ...
    receiverGPSSMass + ...
    droneFrameMass;

% Define general design parameters
gravity = 9.81; % Acceleration due to gravity (m/s^2)
motorThrustCapacity = 1; % Maximum thrust per motor (kgf)
totalThrust = numberOfMotors * motorThrustCapacity;
minimumPayload = 0.5; % Minimum required payload (kg)
requiredThrustToWeightRatio = 2; % Required thrust-to-weight ratio (2:1)
numberOfArms = 4; % Number of arms on the quadcopter

```

Task 2: Propose at least two drone arm designs

Develop at least two distinct drone arm designs by drawing clear, neatly labeled sketches that show key geometric features, including:

- Overall shape (e.g. beam, tube, truss structure)
- Dimensions (length, width, thickness)
- Motor mount location
- Where the arm connects to the drone body

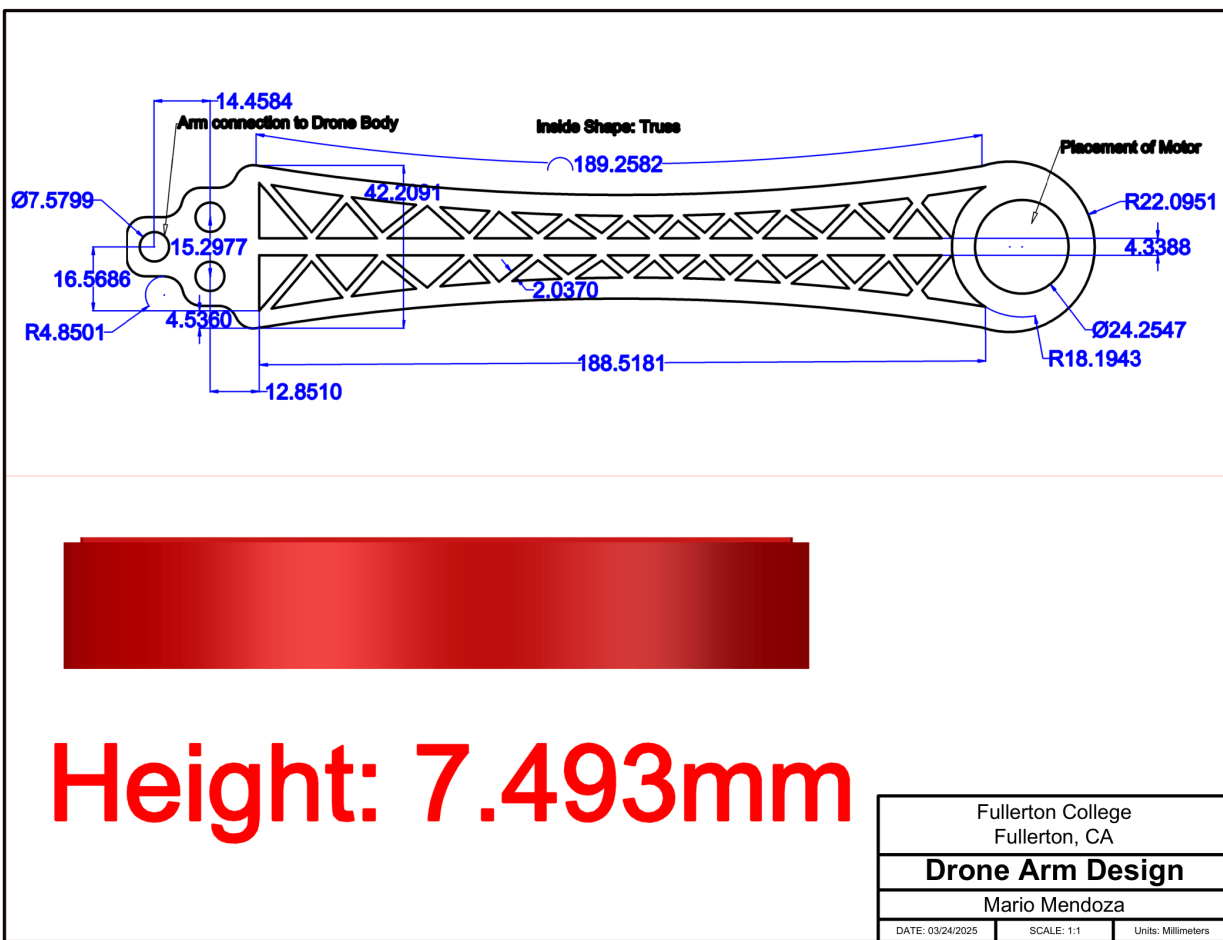
The two designs should be meaningfully different from each other (e.g. different geometries or structural strategies) and not just small variations of the same idea.

Navigate to the 'Insert' tab and insert pictures of your sketches in the space below (hand-drawn or digital). Include a short design description for each sketch that includes:

- Geometry (e.g. solid beam, hollow tube, beam with trusses)
- Why you chose this design
- How it might perform under load

Note: All dimensions and measurements for the drone arm designs are provided in millimeters (mm) for consistency.

Drone Arm Design 1:



Geometry: A hollow, truss-shaped interior where each truss arm or branch has a thickness of approximately 2.0370 mm. The outside walls feature a continuous inward curve (with a radius of 189.2582 mm), causing the internal truss structure to narrow towards the middle before widening again as it approaches the motor housing. Holes for the arm connection are placed in a triangle shape. The total height of the drone arm is 7.493 mm throughout.

Reasoning: Truss shapes of the inside part of the drone arm are designed for stabilization, optimal strength, material efficiency, and weight reduction. This drone arm design also allows for creativity of the user or any future re-designers regarding the placement of the motor. The arm connection to the body of the drone is also shaped in a triangle to ensure the placement is optimally secured and geometrically rigid.

Analysis: While the truss structure of the drone arm design will provide strength and stiffness, the region between the arm connection and the starting portion of where the curvature begins might cause future problems; further optimization might be required in order to avoid a snap of that region.

Task 3: Perform thrust-to-weight analysis across all design and material options

Write a MATLAB function that will perform a thrust-to-weight analysis across all design and material options. This analysis will help you evaluate which drone arm design and which material optimizes payload capacity while meeting suggested safety requirements.

- Your design must support a minimum payload of 0.5kg.
- Your design must meet a thrust-to-weight ratio of at least 2:1 to ensure safe and stable flight. This safety standard is widely used in drone design. Conceptually, it means that the drone's maximum thrust must be at least twice its weight. This ensures stable hovering and maneuvering with a safety margin. Note that the thrust-to-weight safety margin is a system-level safety standard for drone flight that is distinct from the structural safety factor of the drone arms (which you will evaluate later through finite element analysis).
- Your goal is calculate maximum payload capacity for each design and material option (i.e. what is the heaviest payload your drone design could carry while still allowing the drone to safely take off?)

Ensure that the results of your analysis can be readily interpreted, e.g. in the form of a summary table or plot.

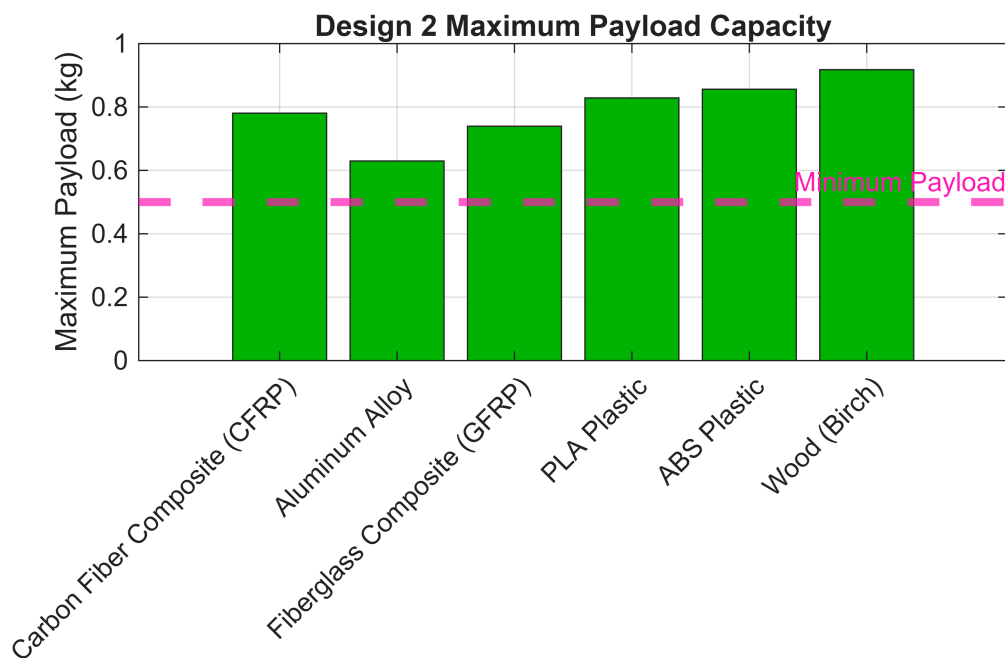
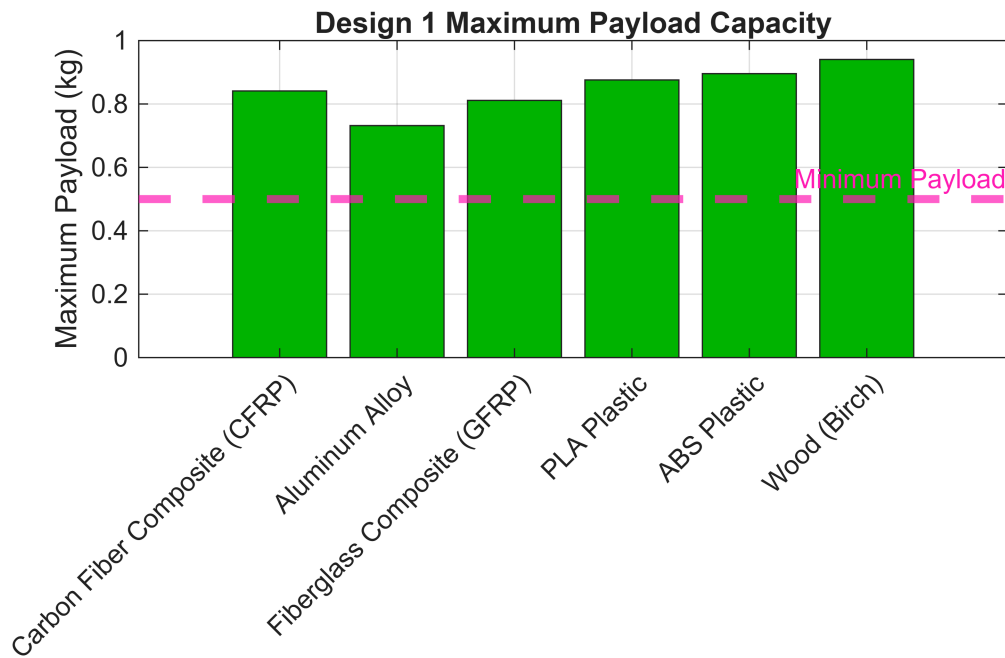
Note: if you have two drone arm designs, the expected output should be 12 payload capacity estimates (2 designs across 6 materials options).

```
% Define arm volumes from Task 2
armVolume1 = 2.483207e-5;
armVolume2 = 3.4295614e-5;

% Store both design volumes for comparison
armVolumes = [armVolume1, armVolume2];

% Define names for each drone arm design
designNames = ["Design 1", "Design 2"];

% Perform thrust-to-weight analysis for all designs and materials
task3Results = thrustToWeightAnalysis( ...
    armVolumes, ...
    designNames, ...
    materials, ...
    totalDroneMassWithoutArmsOrPayload, ...
    totalThrust, ...
    minimumPayload, ...
    requiredThrustToWeightRatio, ...
    numberOfArms);
```



=====

DESIGN 1 MAXIMUM PAYLOAD TABLE

=====

Material	TotalArmMass_kg	MaximumPayload_kg
"Carbon Fiber Composite (CFRP)"	0.15893	0.84107
"Aluminum Alloy"	0.26819	0.73181
"Fiberglass Composite (GFRP)"	0.18872	0.81128
"PLA Plastic"	0.12416	0.87584
"ABS Plastic"	0.10429	0.89571
"Wood (Birch)"	0.059597	0.9404

Design 1 Maximum Payload Summary:

Design 1 Passes for:
 Carbon Fiber Composite (CFRP)
 Aluminum Alloy
 Fiberglass Composite (GFRP)
 PLA Plastic
 ABS Plastic
 Wood (Birch)

Design 1 Fails for:
 No materials failed

=====

DESIGN 2 MAXIMUM PAYLOAD TABLE

=====

Material	TotalArmMass_kg	MaximumPayload_kg
"Carbon Fiber Composite (CFRP)"	0.21949	0.78051
"Aluminum Alloy"	0.37039	0.62961
"Fiberglass Composite (GFRP)"	0.26065	0.73935
"PLA Plastic"	0.17148	0.82852
"ABS Plastic"	0.14404	0.85596
"Wood (Birch)"	0.082309	0.91769

Design 2 Maximum Payload Summary:

Design 2 Passes for:
 Carbon Fiber Composite (CFRP)
 Aluminum Alloy
 Fiberglass Composite (GFRP)
 PLA Plastic
 ABS Plastic
 Wood (Birch)

Design 2 Fails for:
 No materials failed

```
function task3Results = thrustToWeightAnalysis( ...
    armVolumes, designNames, materials, ...
    totalDroneMassWithoutArmsOrPayload, ...
    totalThrust, minimumPayload, ...
    requiredThrustToWeightRatio, ...
    numberOfArms)

% Determine the number of materials and design-material combinations
numberOfMaterials = length(materials);
numberOfCombinations = length(armVolumes) * numberOfMaterials;

% Initialize arrays for storing analysis results
designNameResults = strings(numberOfCombinations, 1);
materialNameResults = strings(numberOfCombinations, 1);
armDensity = zeros(numberOfCombinations, 1);
totalArmMass = zeros(numberOfCombinations, 1);
maximumPayload = zeros(numberOfCombinations, 1);
resultStatus = strings(numberOfCombinations, 1);
```

```

    % Calculate the maximum allowable drone mass based on the required thrust-to-
weight ratio
    maximumAllowedTotalMass = totalThrust / requiredThrustToWeightRatio;

    % Initialize combination counter
    combinationNumber = 1;

    % Loop through each arm design
    for designNumber = 1:length(armVolumes)

        % Get the current arm volume
        currentArmVolume = armVolumes(designNumber);

        % Loop through each material option
        for materialNumber = 1:numberOfMaterials

            % Store current design and material information
            designNameResults(combinationNumber) = designNames(designNumber);
            materialNameResults(combinationNumber) =
string(materials(materialNumber).name);
            armDensity(combinationNumber) = materials(materialNumber).rho_kg_m3;

            % Calculate the total mass of all four drone arms
            totalArmMass(combinationNumber) = ...
                currentArmVolume * armDensity(combinationNumber) * numberOfArms;

            % Calculate the maximum payload capacity for the current design-
material combination
            maximumPayload(combinationNumber) = ...
                maximumAllowedTotalMass - ...
                totalDroneMassWithoutArmsOrPayload - ...
                totalArmMass(combinationNumber);

            % Calculate the total drone mass including the minimum payload
            totalDroneMass = ...
                totalDroneMassWithoutArmsOrPayload ...
                + totalArmMass(combinationNumber) ...
                + minimumPayload;

            % Check if the design satisfies payload and thrust-to-weight
requirements
            if maximumPayload(combinationNumber) >= minimumPayload && ...
                totalThrust / totalDroneMass >= requiredThrustToWeightRatio

                resultStatus(combinationNumber) = "PASS";

            else

                resultStatus(combinationNumber) = "FAIL";

```

```

        end

        % Move to the next design-material combination
        combinationNumber = combinationNumber + 1;

    end
end

% Create a complete table containing all internal analysis results
fullTable = table( ...
    designNameResults, ...
    materialNameResults, ...
    armDensity, ...
    totalArmMass, ...
    maximumPayload, ...
    'VariableNames', { ...
        'Design', ...
        'Material', ...
        'Density_kg_m3', ...
        'TotalArmMass_kg', ...
        'MaximumPayload_kg'});

% Save internal results for later use in Task 5
task3Results.designNames = designNameResults;
task3Results.materialNames = materialNameResults;
task3Results.maximumPayload = maximumPayload;
task3Results.resultStatus = resultStatus;

% Extract display tables for each individual design
design1Display = fullTable( ...
    fullTable.Design == "Design 1", ...
    {'Material', 'TotalArmMass_kg', 'MaximumPayload_kg'});
design2Display = fullTable( ...
    fullTable.Design == "Design 2", ...
    {'Material', 'TotalArmMass_kg', 'MaximumPayload_kg'});

% Create a payload comparison plot for each design
for designNumber = 1:length(designNames)

    % Create a new figure for the current design
    figure;

    % Identify results belonging to the current design
    designIndex = designNameResults == designNames(designNumber);
    designPayload = maximumPayload(designIndex);
    designStatus = resultStatus(designIndex);
    designMaterials = materialNameResults(designIndex);

    % Create bar plot of maximum payload capacity

```

```

barHandle = bar(designPayload);

% Initialize colors based on pass or fail status
plotColors = zeros(numberOfMaterials, 3);

% Assign green to passing materials and red to failing materials
for materialNumber = 1:numberOfMaterials

    if designStatus(materialNumber) == "PASS"
        plotColors(materialNumber, :) = [0 0.7 0];

    else
        plotColors(materialNumber, :) = [0.9 0 0];

    end

end

% Apply colors to the bar chart
barHandle.FaceColor = "flat";
barHandle.CData = plotColors;

% Format x-axis labels
xticks(1:numberOfMaterials);
xticklabels(designMaterials);
xtickangle(45);

% Add plot labels and title
ylabel("Maximum Payload (kg)");
title(designNames(designNumber) + " Maximum Payload Capacity");

% Display minimum payload requirement
yline(minimumPayload, "--", "Minimum Payload", ...
    "Color", [1 0.1 0.7], "LineWidth", 3);

% Set plot limits and enable grid
ylim([0 1]);
yticks(0:0.2:1);
grid on;

end

% Display Design 1 results
fprintf("\n");
fprintf("=====\n");
fprintf("      DESIGN 1 MAXIMUM PAYLOAD TABLE\n");
fprintf("=====\n\n");
disp(design1Display);

% Design 1 Summary:

```

```

fprintf("\nDesign 1 Maximum Payload Summary:\n");

design1Payload = maximumPayload(1:numberOfMaterials);
design1Materials = materialNameResults(1:numberOfMaterials);

% Design 1 passes
fprintf("\nDesign 1 Passes for:\n");

design1Passed = false;

for i = 1:numberOfMaterials
    if design1Payload(i) >= minimumPayload
        fprintf(" %s\n", design1Materials(i));
        design1Passed = true;
    end
end

if ~design1Passed
    fprintf(" No materials passed\n");
end

% Design 1 fails
fprintf("\nDesign 1 Fails for:\n");

design1Failed = false;

for i = 1:numberOfMaterials
    if design1Payload(i) < minimumPayload
        fprintf(" %s\n", design1Materials(i));
        design1Failed = true;
    end
end

if ~design1Failed
    fprintf(" No materials failed\n");
end

% Display Design 2 results
fprintf("\n");
fprintf("===== \n");
fprintf("      DESIGN 2 MAXIMUM PAYLOAD TABLE\n");
fprintf("===== \n\n");
disp(design2Display);

% Design 2 Summary:
fprintf("\nDesign 2 Maximum Payload Summary:\n");

design2Payload = maximumPayload(numberOfMaterials+1:2*numberOfMaterials);
design2Materials = materialNameResults(numberOfMaterials+1:2*numberOfMaterials);

```

```

% Design 2 passes
fprintf("\nDesign 2 Passes for:\n");

design2Passed = false;

for i = 1:numberOfMaterials
    if design2Payload(i) >= minimumPayload
        fprintf(" %s\n", design2Materials(i));
        design2Passed = true;
    end
end

if ~design2Passed
    fprintf(" No materials passed\n");
end

% Design 2 fails
fprintf("\nDesign 2 Fails for:\n");

design2Failed = false;

for i = 1:numberOfMaterials
    if design2Payload(i) < minimumPayload
        fprintf(" %s\n", design2Materials(i));
        design2Failed = true;
    end
end

if ~design2Failed
    fprintf(" No materials failed\n");
end
end

```

Task 4: Perform finite element analysis across all design and material options

Perform finite element analysis (FEA) across all design and material options to evaluate the structural safety factor of your drone arm design. To simplify the task, you will perform the FEA only on a single drone arm.

Using a CAD software of your choice, first create a 3D model of one drone arm for each of your designs. Be sure your model can be exported as a STEP or STL file for analysis in MATLAB.

- Recommended CAD software options: Tinkercad, Onshape, Fusion 360, Solidworks (most have free versions)

Import your model into MATLAB and perform FEA. In your FEA model, you should model and apply two loads to the drone arm: upward thrust force from the motor and downward weight of the motor.

- Helpful resources: [online tutorial](#), documentation for [Structural Mechanics](#) and [Partial Differentiation Equation Toolbox](#), and [example workflow](#)

For the chosen design and each material combination, you should report the overall maximum displacement, maximum stress, and factor of safety and include visualizations of the x-, y-, and z-displacement and Von Mises stress. Ensure that your results are presented for easy interpretation (e.g. present a table with numerical results, clearly labeled subplots for the visualizations).

- The factor of safety compares how strong a material is to how much stress it experiences, indicating whether a design is safe under the given loads. For this project use the equation:
$$\text{FoS} = \frac{\text{Material Yield Strength (Pa)}}{\text{Maximum von Mises Stress}}$$
Yield strength indicates how much stress a given material can withstand before permanently deforming. Maximum von Mises stress (derived from your FEA model) is a measure of how much stress your design experiences under the applied loads.
- A typical engineering safety margin for FoS is ~1.5-2+
- This [example](#) demonstrates the plots you are asked to include in your results

```
% STEP 1: Import drone arm geometry into MATLAB, visualize the geometry to
% identify key faces (e.g. tip face)

% Select which drone arm design to analyze.
% By default, Design 1 will run.
% To run Design 2, comment out the first line below and uncomment the second.
% Ensure the corresponding CAD file is in the same directory as this Live Script.
selectedDesign = "Design 1";
% selectedDesign = "Design 2";

if selectedDesign == "Design 1"

    CADfile = "Design_1_CAD_Model.STEP";
    fixedFaceID = 24;
    motorMountFaceID = 33;
    motorMountFaceArea = 0.00062144;

elseif selectedDesign == "Design 2"

    CADfile = "Design_2_CAD_Model.STL";
    fixedFaceID = 111;
    motorMountFaceID = 82;
    motorMountFaceArea = 0.000866359;

end

% Initialize the finite element model using the geometry
modelFE = femodel(AnalysisType="structuralStatic", Geometry=CADfile);
```

```

% Apply STL scaling only for Design 2
if selectedDesign == "Design 2"
    modelFE.Geometry = scale(modelFE.Geometry,0.001);
end

% Generate the finite element mesh
modelFE = generateMesh(modelFE, Hmax=1.5e-3);

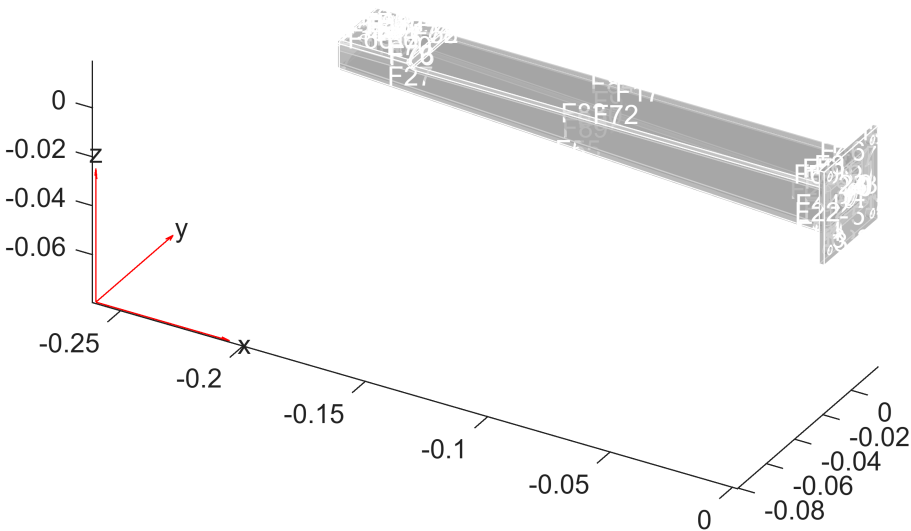
% Plot the geometry and display face labels
% This allows the fixed base and motor mount faces to be identified
figure;

pdegplot(modelFE, FaceLabels="on", FaceAlpha=0.5);

view(30, 30);
axis equal;
title(selectedDesign + " Drone Arm Face Labels");

```

Design 1 Drone Arm Face Labels



```

% IMPORTANT:
% Visually confirm from this face-label plot that the selected face IDs
% correspond to the intended physical locations

% STEP 2: Define material properties

% Get the number of available materials
numberOfMaterials = length(materials);

% Initialize arrays for storing FEA results for all six materials
materialName = strings(numberOfMaterials, 1);

```

```

maximumDisplacementResults = zeros(numberOfMaterials, 1);
maximumVonMisesStressResults = zeros(numberOfMaterials, 1);
factorOfSafetyResults = zeros(numberOfMaterials, 1);
resultStatus = strings(numberOfMaterials, 1);

% STEP 3: Apply boundary conditions (fix base of arm where it would be
% attached to drone body)

modelFE.FaceBC(fixedFaceID) = ...
    faceBC(Constraint="fixed");

% STEP 4: Apply the upward thrust force (motor force)
% Define the thrust force, convert force to surface traction, apply
% surface traction to the appropriate face in the appropriate direction

% Calculate the upward thrust force from the motor
motorThrustForce_N = motorThrustCapacity * gravity;

% STEP 5: Apply downward motor weight
% Define motor weight, convert to traction, apply to the appropriate face
% in the appropriate direction

% Calculate the downward weight of the motor
motorWeightForce_N = motorMass * gravity;

% Calculate the net force acting in the loading direction
% The upward motor thrust is positive and the downward motor weight is
% negative
netForceZ_N = motorThrustForce_N - motorWeightForce_N;

% IMPORTANT:
% SurfaceTraction is specified in the global Cartesian coordinate system
% The current model assumes that positive global Z is the upward thrust
% direction based on the STEP model coordinate orientation

thrustDirection = [0; 0; 1];

% Calculate the net surface traction
netTractionMagnitude = netForceZ_N / motorMountFaceArea;

% Convert the scalar net traction into a global Cartesian vector
netTraction = thrustDirection * netTractionMagnitude;

% Apply the combined net traction to the motor mount face
modelFE.FaceLoad(motorMountFaceID) = faceLoad(SurfaceTraction=netTraction);

% STEP 6: Solve the FEA model

```

```

for materialNumber = 1:numberOfMaterials

    % Print the current material being processed
    fprintf("Running FEA for material %d/%d: %s\n", ...
        materialNumber, numberOfMaterials, string(materials(materialNumber).name));
    drawnow;

    % Get material information from the droneArmMaterials data structure
    materialName(materialNumber) = string(materials(materialNumber).name);

    YoungsModulus = materials(materialNumber).E_Pa;
    PoissonsRatio = materials(materialNumber).nu;
    YieldStrength = materials(materialNumber).yieldStrength_Pa;

    % Assign material properties to the finite element model
    currentModel = modelFE;

    currentModel.MaterialProperties = materialProperties( ...
        YoungsModulus=YoungsModulus, ...
        PoissonsRatio=PoissonsRatio);

    % Solve the FEA model for the current material
    results = solve(currentModel);

    % STEP 7: Get numerical results

    % Calculate displacement magnitude at every mesh node
    displacementMagnitude = sqrt( ...
        results.Displacement.ux.^2 + ...
        results.Displacement.uy.^2 + ...
        results.Displacement.uz.^2);

    % Calculate the maximum displacement magnitude
    maximumDisplacement = max(displacementMagnitude);

    % Calculate the maximum Von Mises stress
    maximumVonMisesStress = max(results.VonMisesStress);

    % Calculate the factor of safety
    factorOfSafety = YieldStrength / maximumVonMisesStress;

    % Store numerical results (conversion of units)
    maximumDisplacementResults(materialNumber) = maximumDisplacement * 1000;
    maximumVonMisesStressResults(materialNumber) = maximumVonMisesStress / 1e6;
    factorOfSafetyResults(materialNumber) = factorOfSafety;

    % Determine whether the design meets the recommended
    % minimum factor of safety of 1.5
    if factorOfSafety >= 1.5

```

```

        resultStatus(materialNumber) = "SAFE";
    else
        resultStatus(materialNumber) = "UNSAFE";
    end

% STEP 8: Visualize your results
figure;
sgtitle(selectedDesign + " - " + materialName(materialNumber) + " FEA Results");

subplot(2,2,1);
pdeplot3D(results.Mesh, ColorMapData=results.Displacement.ux);
title("X Displacement");
xlabel("X");
ylabel("Y");
zlabel("Z");
axis equal;
colormap(gca, "jet");
colorbar;

subplot(2,2,2);
pdeplot3D(results.Mesh, ColorMapData=results.Displacement.uy);
title("Y Displacement");
xlabel("X");
ylabel("Y");
zlabel("Z");
axis equal;
colormap(gca, "jet");
colorbar;

subplot(2,2,3);
pdeplot3D(results.Mesh, ColorMapData=results.Displacement.uz);
title("Z Displacement");
xlabel("X");
ylabel("Y");
zlabel("Z");
axis equal;
colormap(gca, "jet");
colorbar;

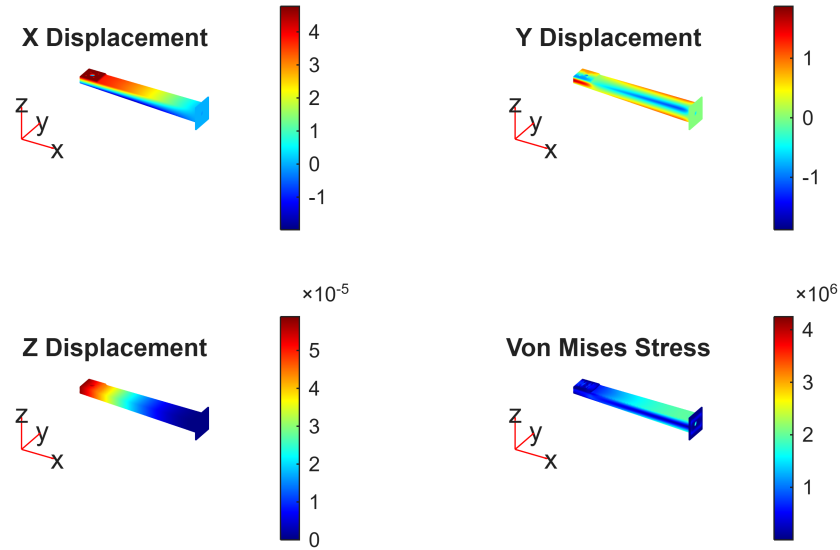
subplot(2,2,4);
pdeplot3D(results.Mesh, ColorMapData=results.VonMisesStress);
title("Von Mises Stress");
xlabel("X");
ylabel("Y");
zlabel("Z");
axis equal;
colormap(gca, "jet");
colorbar;

end

```

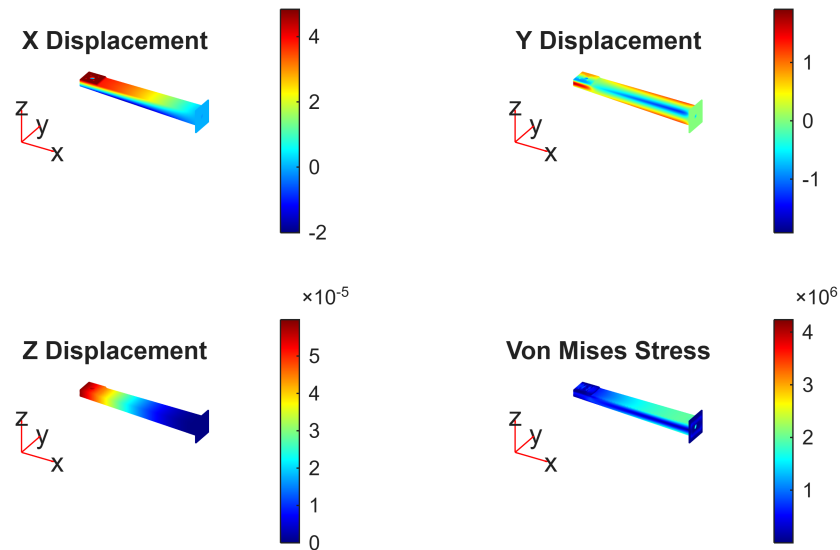
Running FEA for material 1/6: Carbon Fiber Composite (CFRP)

Design 1 - Carbon Fiber Composite (CFRP) FEA Results



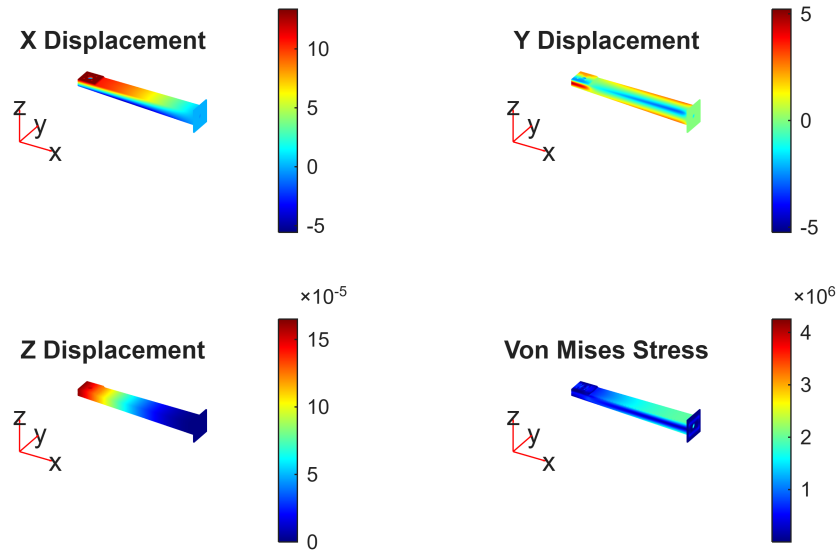
Running FEA for material 2/6: Aluminum Alloy

Design 1 - Aluminum Alloy FEA Results



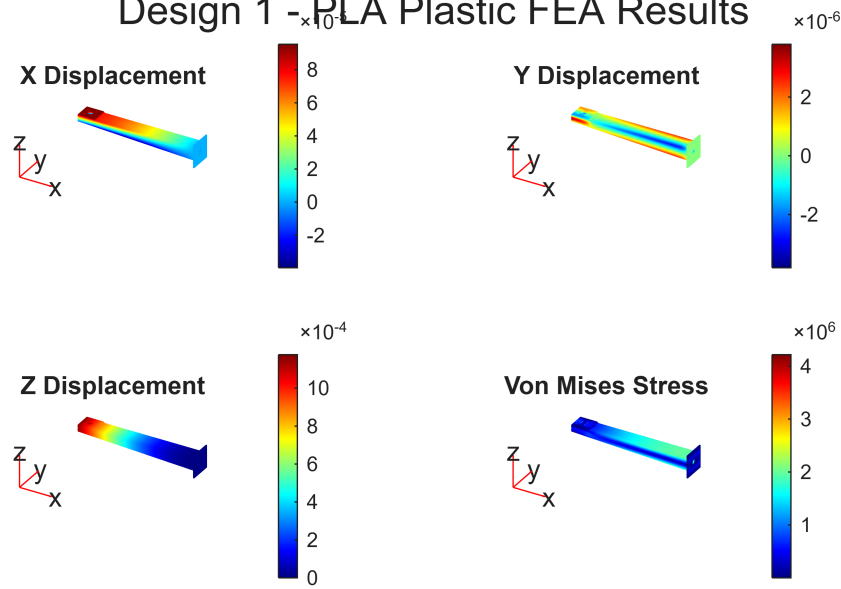
Running FEA for material 3/6: Fiberglass Composite (GFRP)

Design 1 - Fiberglass Composite (GFRP) FEA Results



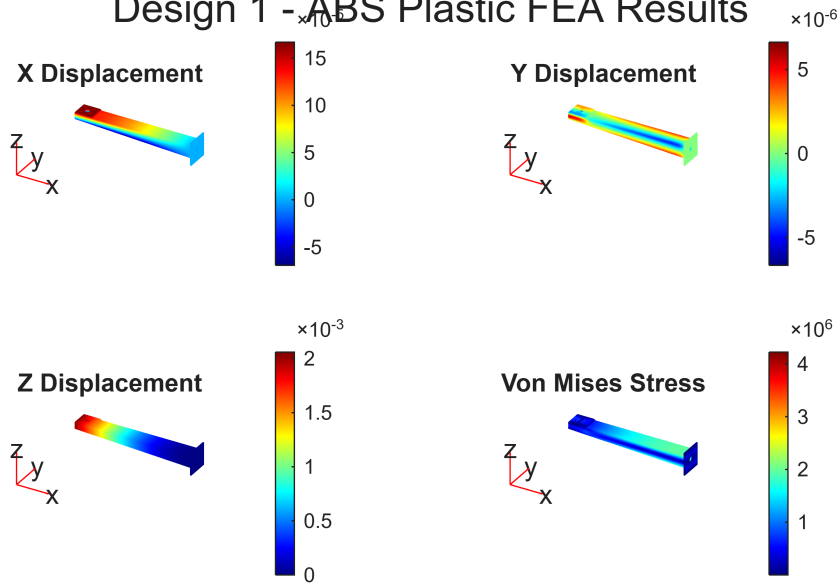
Running FEA for material 4/6: PLA Plastic

Design 1 - PLA Plastic FEA Results



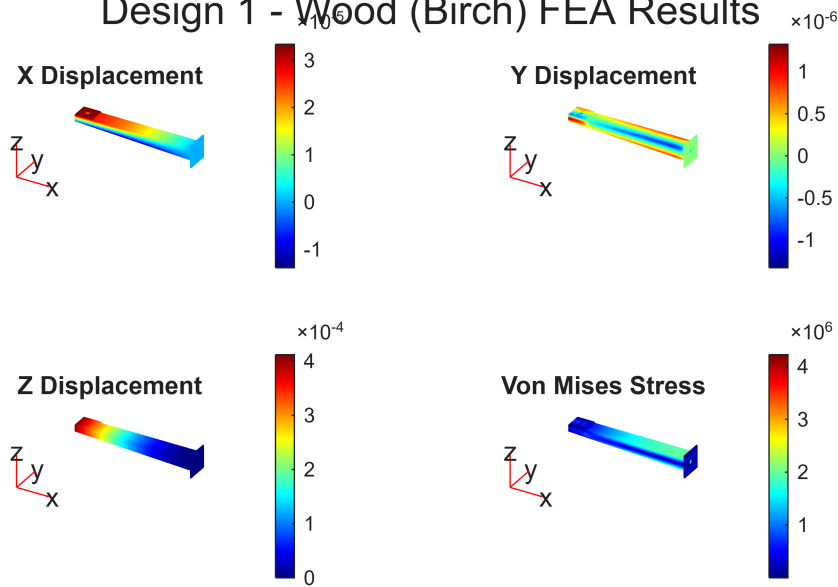
Running FEA for material 5/6: ABS Plastic

Design 1 - ABS Plastic FEA Results



Running FEA for material 6/6: Wood (Birch)

Design 1 - Wood (Birch) FEA Results



% Create a summary table containing the key FEA results for all six materials

```
feaResultsTable = table( ...
    materialName, ...
    maximumDisplacementResults, ...
    maximumVonMisesStressResults, ...
    factorOfSafetyResults, ...
    'VariableNames', { ...
        'Material', ...
        'MaximumDisplacement_mm', ...
        'MaximumVonMisesStress_MPa', ...
```

```
'FactorOfSafety'}));
```

```
% Save complete internal results for later use in Task 5
```

```
task4Results.materialNames = materialName;
```

```
task4Results.factorOfSafety = factorOfSafetyResults;
```

```
task4Results.resultStatus = resultStatus;
```

```
% Display the FEA summary table
```

```
fprintf("\n===== \n          %s FEA RESULTS\nTABLE\n===== \n\n", ...  
        upper(selectedDesign));
```

```
=====  
        DESIGN 1 FEA RESULTS TABLE  
=====
```

```
disp(feaResultsTable);
```

Material	MaximumDisplacement_mm	MaximumVonMisesStress_MPa	FactorOfSafety
"Carbon Fiber Composite (CFRP)"	0.059081	4.2484	141.23
"Aluminum Alloy"	0.059894	4.2349	64.937
"Fiberglass Composite (GFRP)"	0.1655	4.2546	70.512
"PLA Plastic"	1.1798	4.2166	14.229
"ABS Plastic"	2.0652	4.2232	9.4714
"Wood (Birch)"	0.41304	4.2232	18.943

```
% Display FEA safety summary
```

```
fprintf("\n%s FEA Safety Summary:\n\nMaterials passing FEA requirements:\n", ...  
        selectedDesign);
```

Design 1 FEA Safety Summary:

Materials passing FEA requirements:

```
feaPassed = false;
```

```
for i = 1:numberOfMaterials
```

```
    if resultStatus(i) == "SAFE"
```

```
        fprintf(" %s\n", materialName(i));
```

```
        feaPassed = true;
```

```
    end
```

```
end
```

```
Carbon Fiber Composite (CFRP)  
Aluminum Alloy  
Fiberglass Composite (GFRP)  
PLA Plastic  
ABS Plastic  
Wood (Birch)
```

```

if ~feaPassed
    fprintf(" No materials passed\n");
end

fprintf("\nMaterials failing FEA requirements:\n");

```

Materials failing FEA requirements:

```

feaFailed = false;

for i = 1:numberOfMaterials
    if resultStatus(i) == "UNSAFE"
        fprintf(" %s\n", materialName(i));
        feaFailed = true;
    end
end

if ~feaFailed
    fprintf(" No materials failed\n");
end

```

No materials failed

Task 5: Optimize the total material cost of the drone arm

Using the results from the thrust-to-weight analysis and finite element analysis, determine which material combination provides the lowest material cost while still meeting the required payload, thrust-to-weight ratio, and Factor of Safety.

Calculate the total material cost for each material option using the average material cost provided in the droneArmMaterials.mat file. Compare the valid material options and identify the option that provides the best balance of cost, performance, and structural safety.

```

% Number of materials
numberOfMaterials = length(materials);

% Initialize arrays
materialName = strings(numberOfMaterials, 1);
totalMaterialCost = zeros(numberOfMaterials, 1);
maximumPayload = zeros(numberOfMaterials, 1);
factorOfSafety = zeros(numberOfMaterials, 1);

task3Status = strings(numberOfMaterials, 1);
task4Status = strings(numberOfMaterials, 1);
finalStatus = strings(numberOfMaterials, 1);

% Select design evaluated in Task 5

if selectedDesign == "Design 1"

```

```

    armLength = 0.202;

elseif selectedDesign == "Design 2"

    armLength = 0.251;

end

% Loop through materials
for materialNumber = 1:numberOfMaterials

    materialName(materialNumber) = string(materials(materialNumber).name);

    % Calculate total material cost for all four arms (rounded to the
    % nearest cent)
    totalMaterialCost(materialNumber) = round( ...
        materials(materialNumber).cost_USD_per_m ...
        * armLength ...
        * numberOfArms, 2);

    % Find selected design result from Task 3
    designIndex = ...
        task3Results.designNames == selectedDesign & ...
        task3Results.materialNames == materialName(materialNumber);

    % Store Task 3 numerical result
    maximumPayload(materialNumber) = ...
        task3Results.maximumPayload(designIndex);

    % Store Task 3 status internally
    task3Status(materialNumber) = ...
        task3Results.resultStatus(designIndex);

    % Find material result from Task 4
    materialIndex = ...
        task4Results.materialNames == materialName(materialNumber);

    % Store Task 4 numerical results
    factorOfSafety(materialNumber) = ...
        task4Results.factorOfSafety(materialIndex);

    % Store Task 4 status internally
    task4Status(materialNumber) = ...
        task4Results.resultStatus(materialIndex);

    % Final design decision requires both tasks to pass
    if task3Status(materialNumber) == "PASS" && ...
        task4Status(materialNumber) == "SAFE"

        finalStatus(materialNumber) = "PASS";
    end
end

```

```

else

    finalStatus(materialNumber) = "FAIL";

end

end

% Create optimization summary table
optimizationResults = table( ...
    materialName, ...
    totalMaterialCost, ...
    maximumPayload, ...
    factorOfSafety, ...
    'VariableNames', { ...
        'Material', ...
        'TotalMaterialCost_USD', ...
        'MaximumPayload_kg', ...
        'FactorOfSafety'});

% Plot cost comparison
figure;

barHandle = bar(totalMaterialCost);

plotColors = zeros(numberOfMaterials, 3);

for materialNumber = 1:numberOfMaterials

    if finalStatus(materialNumber) == "PASS"

        plotColors(materialNumber, :) = [0 0.7 0];

    else

        plotColors(materialNumber, :) = [0.9 0 0];

    end

end

barHandle.FaceColor = "flat";
barHandle.CData = plotColors;

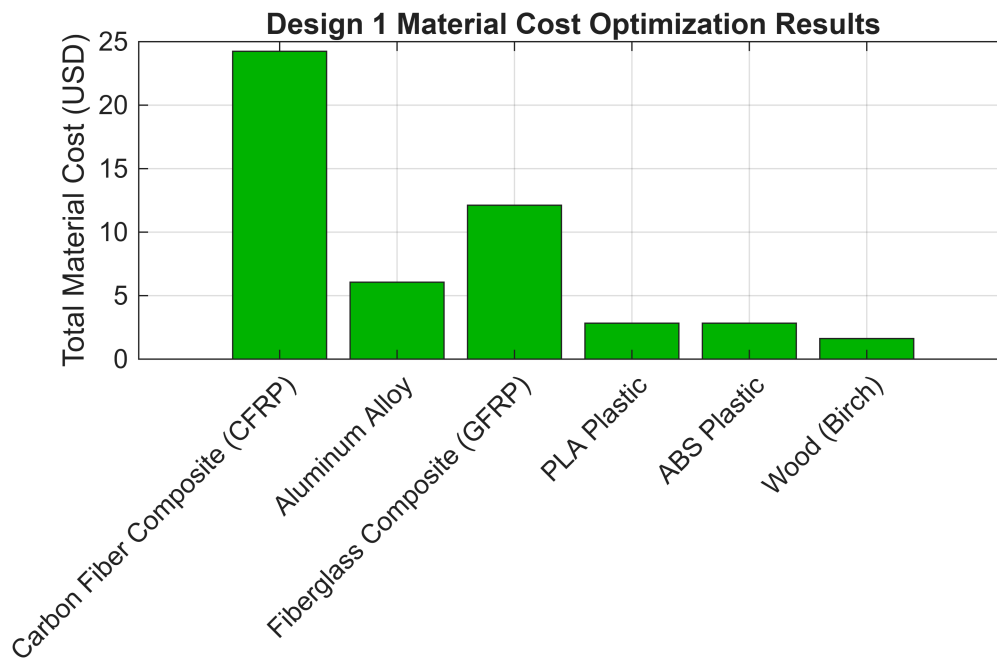
xticks(1:numberOfMaterials);
xticklabels(materialName);
xtickangle(45);

ylabel("Total Material Cost (USD)");

```

```
title(selectedDesign + " Material Cost Optimization Results");
```

```
grid on;
```



```
% Display numerical optimization results
```

```
fprintf("\n===== \n %s MATERIAL OPTIMIZATION\n\n", ...
upper(selectedDesign));
```

```
=====
DESIGN 1 MATERIAL OPTIMIZATION TABLE
=====
```

```
disp(optimizationResults);
```

Material	TotalMaterialCost_USD	MaximumPayload_kg	FactorOfSafety
"Carbon Fiber Composite (CFRP)"	24.24	0.84107	141.23
"Aluminum Alloy"	6.06	0.73181	64.937
"Fiberglass Composite (GFRP)"	12.12	0.81128	70.512
"PLA Plastic"	2.83	0.87584	14.229
"ABS Plastic"	2.83	0.89571	9.4714
"Wood (Birch)"	1.62	0.9404	18.943

```
% Find lowest cost material that satisfies all requirements
```

```
validOptions = optimizationResults(finalStatus == "PASS", :);
```

```
if isempty(validOptions)
```

```
fprintf("\nNo material option satisfies all required constraints.\n");
```

```
else
```

```
[lowestCost, lowestIndex] = min(validOptions.TotalMaterialCost_USD);
```

```
fprintf("\nLowest cost material satisfying all requirements:\nMaterial: %s\nMaterial Cost: %.2f USD\n", ...  
    validOptions.Material(lowestIndex), lowestCost);
```

```
end
```

```
Lowest cost material satisfying all requirements:  
Material: Wood (Birch)  
Material Cost: 1.62 USD
```

Interpretation of Results - Final Recommended Design Solution

Using the results of your thrust-to-weight analysis and finite element analysis, recommend a final drone arm design that meets all required constraints. Your recommendation should clearly identify the selected design (arm geometry and material choice), make use of quantitative results to justify your decisions, demonstrate that the design meets the constraints, explain the trade-offs considered, and include comparisons to alternative designs.

What are some limitations of your work?

What are practical next steps?

The final recommended drone arm design that meets all required constraints is **Design 1**, the hollow-tapered rectangular tube. The final recommended material choice for this design is **Fiberglass Composite (GFRP)**. This recommendation was based not only on quantitative results, such as **maximum payload capacity**, **maximum displacement**, **factor of safety (FoS)**, and **cost**, but also qualitative feasibility such as manufacturability and real-world material nuances that cannot be captured through this MATLAB analysis.

Before beginning the analysis, it is important to contextualize the geometry of both designs. **Design 1** is a hollow-tapered rectangular tube structure that strategically allocates material where it is most needed. Since the arm experiences the highest bending loads near the connection point to the main drone body, **Design 1** was designed with larger cross-sectional dimensions and thicker walls near the base while reducing material toward the motor mount, where less structural reinforcement is required. This approach allowed **Design 1** to maintain the necessary structural capability while minimizing unnecessary material usage. In comparison, **Design 2** uses a hollow truss-shaped internal structure intended to increase rigidity while reducing mass. However, a fundamental limitation was hypothesized (and later proved) to make it less effective: the horizontal orientation of the truss members made them less effective at resisting the dominant vertical bending loads generated by motor thrust. Therefore, prior to the analysis, it was surmised that **Design 1** provided a more practical geometric solution.

The first constraint is that the design must support a minimum payload of **0.5 kg** and achieve a **thrust-to-weight ratio of at least 2:1** to ensure safe and stable flight. Given that the quadcopter has four motors, each producing **one kilogram of thrust**, the maximum allowable weight of the entire quadcopter is **two kilograms**. Since one kilogram is reserved for components of the drone other than the drone arms (i.e., frame, battery, motors, etc.), this leaves a weight budget of **one kilogram** for the remaining four drone arms and the minimum

payload. Based on the drone design sketches and subsequent CAD volumes, it was determined that both designs successfully passed all the requirements. More importantly, the findings indicated that **Design 1** had a smaller volume than **Design 2**, approximately $2.483207 \times 10^{-5} \text{ m}^3$ (**24.83 cm³**) compared to $3.4295614 \times 10^{-5} \text{ m}^3$ (**34.30 cm³**), and thus had less weight overall. This provided further indication that **Design 1** was the better option, both because of its strategic use of geometry relative to the limitations identified in **Design 2**, and its better thrust-to-weight performance. As for the materials, **GFRP** used in **Design 1** allowed for a **maximum payload capacity of 0.81128 kilograms**. However, this was not the highest-performing material in this respect, as **CFRP, PLA, ABS, and Wood** all produced higher maximum payload capacities of **0.84107 kg, 0.87584 kg, 0.89571 kg, and 0.94040 kg**, respectively. Only **Aluminum Alloy** performed worse, producing a **maximum payload capacity of 0.73181 kilograms**.

The second requirement for the drone arm design was that the engineering safety margin, represented by the **Factor of Safety (FoS)**, must be **~1.5-2+** under the given loading conditions of the motor applied to the motor mount face, while a separate face on the root attachment flange was fixed to represent the attachment point to the main drone body. Although this is a simplified representation of the actual loading conditions experienced during flight, it still produced meaningful results for comparing the structural performance of each material. All materials passed the required safety criteria. In particular, **GFRP** produced a **maximum Von Mises stress of 4.2546 MPa**, a **maximum displacement of 0.1655 mm**, and a **Factor of Safety of 70.512**. However, **GFRP** was still not the highest-performing option. **CFRP** produced the **lowest maximum displacement of 0.059081 mm** and the **highest Factor of Safety of 141.23**, while **Aluminum Alloy** produced a **maximum displacement of 0.059894 mm** and a **Factor of Safety of 64.937**. **Wood** produced a **higher maximum displacement of 0.41304 mm** and a **Factor of Safety of 18.943**. The largest concern appeared with the plastic materials, specifically **PLA** and **ABS**. While both materials technically passed the **FoS** requirement, **PLA** produced a **maximum displacement of 1.1798 mm** with a **Factor of Safety of 14.229**, and **ABS** produced a **maximum displacement of 2.0652 mm** with a **Factor of Safety of 9.4714**. These displacement values may seem reasonable, however, when compared to the actual wall thickness of **Design 1**, which tapers down to approximately **1.0 mm** near the motor mount, the issue becomes apparent. The motor thrust can not be assumed to be aligned with the vertical loading direction because the displacement is greater than twice the thickness of the thinnest section of the arm. In real world conditions, such large deformation would introduce vibrations and dynamic instability, referred to in drone applications as the **"jello effect."** Although **PLA** and **ABS** satisfy the numerical **FoS** requirement, their excessive deformation makes them more likely to fail as material choices for this application.

Finally, a **cost optimization** was performed to determine which material satisfied all numerical requirements while minimizing the total material cost. Although **GFRP** resulted in a **total material cost of \$12.12**, it was not the least expensive option. **Wood** produced the **lowest material cost at only \$1.62**, followed by **PLA** and **ABS** at **\$2.83**, **Aluminum Alloy** at **\$6.06**, **GFRP** at **\$12.12**, and **CFRP** at **\$24.24**. Based solely on the numerical optimization, **Wood** would therefore be selected as the optimal material since it also achieved the **highest maximum payload capacity of 0.94040 kg** while maintaining a **Factor of Safety of 18.943**. However, this optimization does not capture several important real-world engineering considerations. Since **Wood** is a natural material, its mechanical properties are not consistent due to grain orientation, knots, moisture absorption, and manufacturing inconsistencies. Consequently, despite its excellent numerical performance and exceptionally low cost, **Wood** was not considered the most practical material for a drone arm. At the other end of the spectrum, **CFRP** provided the highest structural performance but was ultimately

deemed unnecessary when **GFRP** produced similarly strong results for nearly half the cost. In conclusion, **Design 1** manufactured from **GFRP** provides the most balanced engineering solution by satisfying all design requirements while meeting an effective balance between structural performance, manufacturing practicality, material consistency, and overall cost.

Several limitations should be considered when interpreting these results. The accuracy of the finite element analysis depends on mesh size; therefore, a reasonably refined mesh that demonstrated satisfactory convergence of the results was used, although a more comprehensive mesh convergence study with further refinement should be performed in future work. The **FEA** also only analyzed a single drone arm rather than the complete drone assembly, meaning interactions between the arms, central frame, fasteners, and motor mounting hardware were not included. In addition, the material models assumed **isotropic behavior**, which simplifies the actual behavior of composite materials such as **GFRP**, whose properties depend on fiber orientation and manufacturing processes. The applied loading conditions were also simplified by representing the motor thrust and motor weight as a single surface traction, without accounting for real flight effects such as vibration, acceleration, impact loading, aerodynamic forces, or changing thrust conditions. The calculated **Factor of Safety** did not account for fatigue, manufacturing defects, environmental effects, or long-term material degradation.

For future steps, the drone assembly will be tested using more realistic loading conditions and **material properties**, while validating the numerical results through physical prototype testing.